UNIT II

Chapter 3

Tissue Level of Organisation

Chapter Outline

3.1 Animal Tissues
3.2 Epithelial Tissue
3.3 Connective Tissue
3.4 Muscle Tissue
3.5 Neural Tissue



A reflective layer of tissue called tapetum lucidum, enhances night time vision in most of the animals like cat.

Learning Objectives:

- *Recognises the types of tissues based on their characteristic features*
- *Understands the description, location, functions and modification of tissues.*
- *Understands the significance of muscles, connective and neural tissues.*

In multicellular organisms, cells do not operate independently, instead, they form tight cell communities that live and work together. Individual body cells are specialized, with each type performing specific functions that helps to maintain homeostasis and benefits the body as a whole. Cell specialization is obvious. How the muscle cell looks and acts differs greatly from skin cells. Cell specialization allows the body to function in co-ordinated ways. Groups of cells that are similar in structure and perform common or related functions are called '**tissues**'.

Tissues are organized in specific proportions and patterns to form **organs** like lungs, heart, stomach, kidneys, ovaries, testes etc; hence the tissues are called the '**living fabrics**'. If two or more organs perform common physical and chemical functions they are called '**organ systems**', Eg: digestive system, respiratory system, circulatory system, excretory system, etc. Most organs contain different types of tissues and their arrangement determines the organs structure and functions. The study of tissues is called **histology** complements the study of gross anatomy. Together they provide the structural basis for understanding organ physiology.

3.1 Animal Tissues

Animal tissues are classified according to the size, shape and function of the cells. There are four primary (basic) tissue types that interweave to form the 'fabric' of the body. They are, the epithelial tissue (covering), the connective tissue (support), the muscle tissue (movement) and the nervous tissue (control) (Figure 3.1).

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